

Snowflake CoCo CLI Enterprise SupplyGuard Agent

Category: [Intelligent Workflow Automation Agents](#) | Team: [EquiSaaS BD](#)

Executive Overview:

Global supply chains suffer millions in losses due to unexpected shipping delays, vendor lead-time anomalies, and unmonitored stockout risks. Traditional BI tools only alert after line stoppage occurs.

Our solution, **Cortex-SupplyGuard**, leverages **Snowflake CoCo CLI Agent Skills** to autonomously scan enterprise warehouse data, synthesize unstructured incident logs using Snowflake Cortex AI, and execute policy-bounded transactional mitigations in seconds.

Key Business Metric	Traditional Manual Process	Autonomous CoCo Agent
Resolution Time (MTTR)	2 - 5 Days of manual triage	Instant (< 3 Seconds)
Stockout Prevention	Reactive after production halt	100% Proactive Buffer Reroute
Quantified Risk Avoided	High financial leakage	\$200,000+ per incident
Governance & Policy	Prone to manual oversight	100% Auditable Ledger

Modular CoCo Agent Skills Architecture

Built using standardized SKILL.md CoCo specification format

Skill Name & Directory	Role & Capabilities	Input / Output Schema
data-anomaly-detector
Snowflake CoCo Skills/data_anomaly_detector	Autonomous Data Anomaly Detection • Monitors data streams for anomalies in real-time • Detects unusual patterns, trends, and outliers • Alerts on anomalies via email, Slack, or PagerDuty	Input: Data Stream (Table/View) Output: Anomaly Report (JSON)
cortex-reasoner
Snowflake CoCo Skills/cortex_reasoner	Contextual Reasoning • Synthesizes unstructured data into actionable insights • Identifies trends, correlations, and anomalies • Provides recommendations for data-driven decisions	Input: Structured Data (Table/View) Output: Reasoning Log (JSON)
action-executor
Snowflake CoCo Skills/action_executor	Policy Enforcement • Enforces security policies and access controls • Issues emergency actions (e.g., revoke, drop) • Audits actions and logs for compliance	Input: Policy Rules (JSON) Output: Action Log (JSON)

Multi-Step Orchestration & User Interfaces

Stateful execution engine, interactive CLI, and Glassmorphism Web Dashboard

Stateful Workflow Execution Flow:

- 1. **Data Ingestion:** Skill 1 scans Snowflake warehouse inventory for critical lead-time anomalies.
- 2. **Cortex Reasoning:** Skill 2 fuses metrics with unstructured logistics logs via Snowflake Cortex AI to synthesize root causes.
- 3. **Decision Branching:** Policy engine evaluates budget thresholds (Auto-Approve < \$50,000 vs Human Escalation >= \$50,000).
- 4. **Action & Audit:** Skill 3 issues emergency POs, updates warehouse buffer stock, and records audit trail.

Dual User Interfaces Included:

Interface Mode	Features & Capability	Execution Command
Rich Interactive CLI	ANSI tables, step progress bars, trace logs, interactive query prompt	<code>python main.py --demo</code> / <code>run_
Glassmorphism Web UI	Live metric cards, interactive DAG animation, real-time reasoning console, audit ledger	<code>python main.py --web</code> / <code>run_